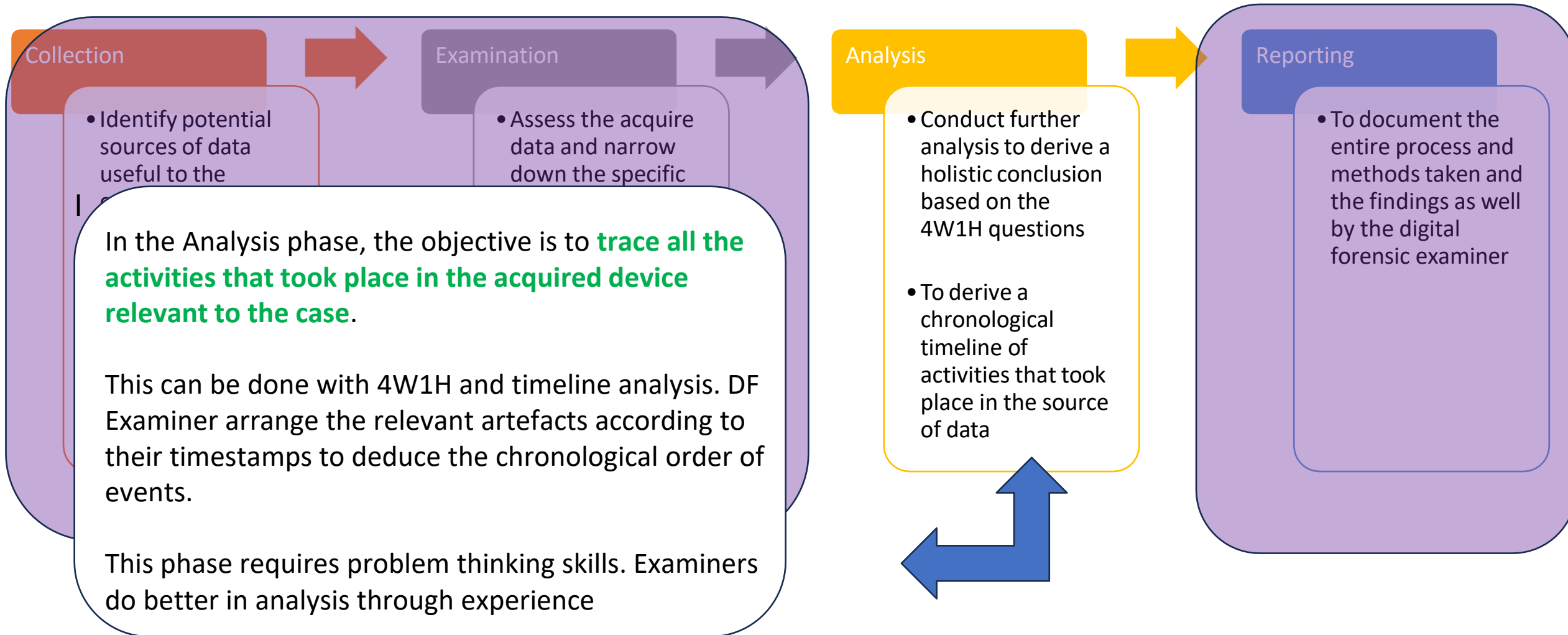


Week 5

Digital Forensic Framework: Analysis

Digital Forensic Framework by NIST



Analysis

Temporal

- Entails a chronological timeline of events that happened among the people, devices and locations involved in the case.

Relational

- Determining the relationships between the people and devices involved in a case through visual descriptions.
- Create visual connections to reveal the patterns in the case.

Functional

- Determining the involved people's behaviours and devices' capability to partake in the crime.

Timeline Analysis (Temporal)

- Timeline is the capability that allows a DF examiner to automatically assemble artefacts from various file paths in temporal order.
- The timeline can then be examined to see the interaction of many of these items. **This method has proven to speed up analytical work.**
- Upon the creating the timeline, it still relies on the DF examiner's skills to infer the meaning of events to be able to answer the 4W1H questions (What, Why, Where, Who and How).

Functional Analysis

- To solve a crime involving digital evidence, a DF examiner must understand the **unique behavioural characteristics** of both offender and digital evidence.

Offender	1. Is the offender a verbally abusive person? 2. Does the offender have records of stalking others before? 3. Does the offender have the tendency to implicate physical harm on others?
Digital Evidence	1. Does this mobile device have the functionality to send WhatsApp messages? 2. Does this mobile device have the functionality to store and run dual SIM cards? 3. Does this mobile device have the functionality to store media files in hidden folders?

Functional Analysis

- For example, in a case of Harassment, the behaviour of the offender can be identified from their communication exchange with the victim.
- The language and tone used by the offender may reveal his state of mind during his exchange with the victim.
 - In this screenshot example, the messages sent by the offender indicates **anger and dissatisfaction**.
 - It further indicates that the offender is **verbally abusive**.
 - The messages sent by him prove that he can commit a crime of harassment.



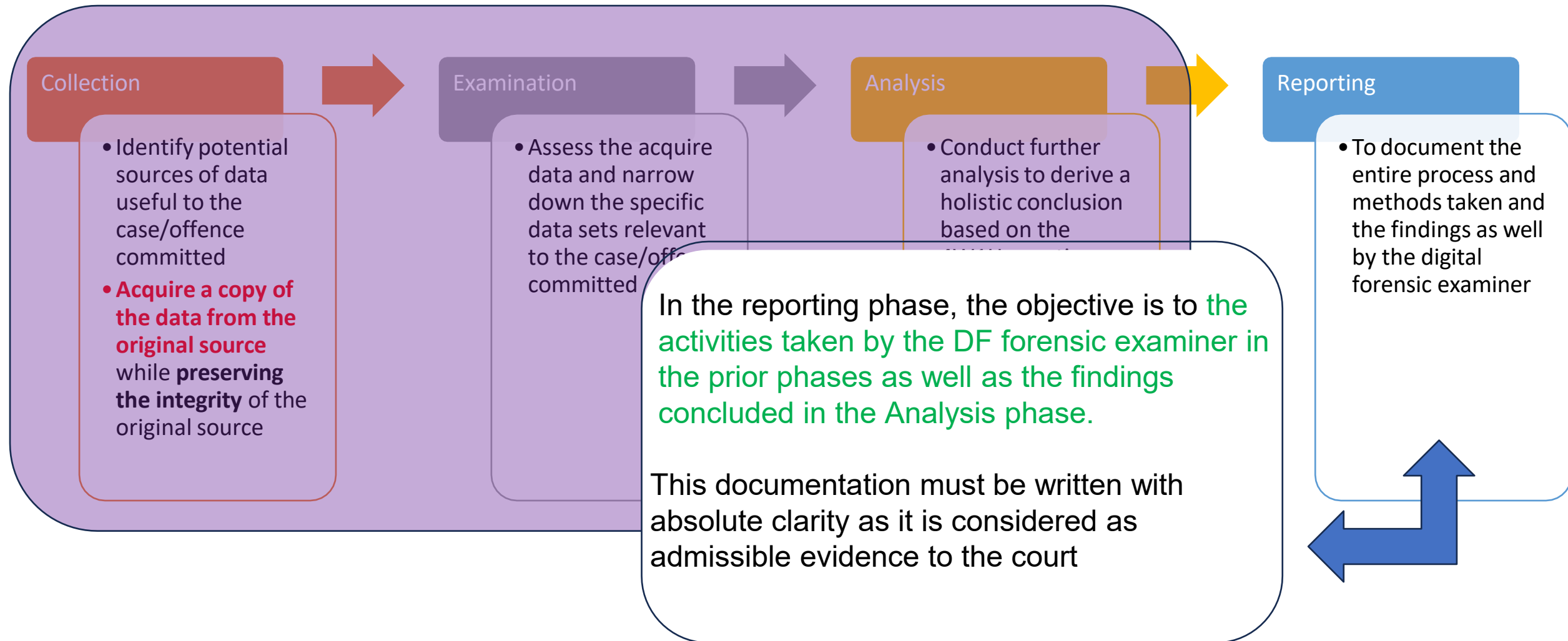
Functional Analysis

- In this same case scenario, we can also uniquely identify that the mobile device used by the offender have the capability to partake in the crime because it is able to run WhatsApp application.
- We can testify that because **this mobile device is running the latest Android/IOS version**, it can send and receive messages via WhatsApp, a third-party messaging application.



Digital Forensic Framework: Reporting

Digital Forensic Framework by NIST



Reporting

- Reporting is the process of preparing a **detailed summary of all steps taken and conclusions** reached as part the examination and analysis. Reporting should include details of all important actions performed by the DF examiner, the results of the acquisition and any inferences drawn from the results.
- The report written by the DF examiner can be submitted to court as evidence.

Reporting

- A well rounded report may contain (not limited to) the following details:
 - Identity of the DF examiners involved in the case
 - Nature of the case (What is the offence committed)
 - Objective of the case analysis
 - Methods taken to preserve the evidence at crime scenes
 - Chain of Custody
 - Information of the devices found (Make and Model, Serial Number, external damages like damaged screen)
 - Forensic tools and/or methods taken during acquisition of evidence
 - Description of steps taken during examination and analysis
 - Findings related to the evidence (messages, media files, emails, geolocations etc)
 - Conclusion

Learning Topics in Week 6

- Android Architecture
- Android Security: Trusted Execution Environment (TEE)
- Android Security: Full Disk Encryption (FDE)
- Android Security: File Based Encryption (FBE)